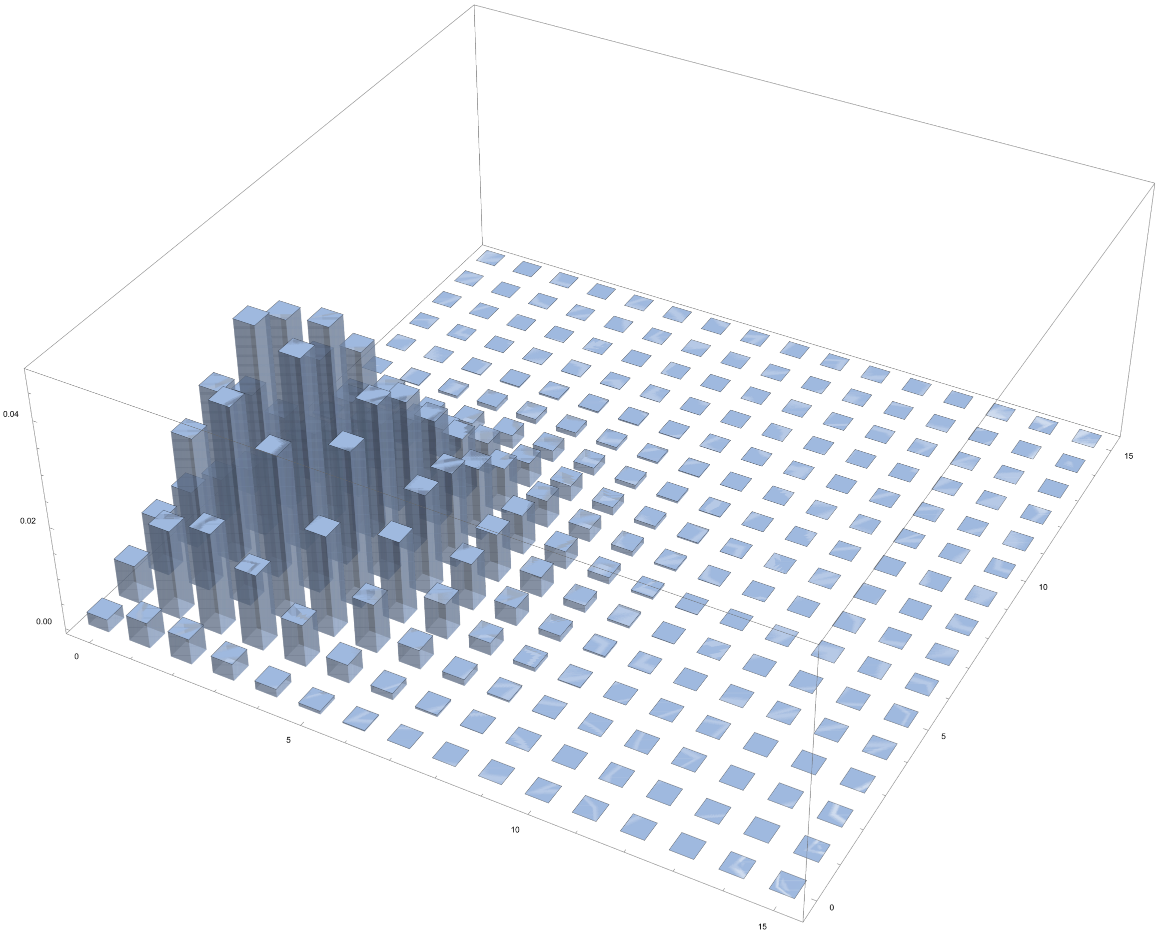

Probability and Combinatorics

The Stochastic Art of Throwing Dice

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Weakly Descending

How many 4-digit numbers have their digits in weakly descending order? In these numbers, every digit beginning with the hundreds digit is less than or equal to the digit before it.

SOLUTION (Longer):

Consider the following scenario. We have, in a row, 9 boxes with labels in descending order from 9 to 1. We also have 4 identical balls. It is key to realize that the number of combinations of placing the balls in the boxes is synonymous with the original question. If each ball is a digit and corresponds to the digit of the box it is in, then that combination is a unit ordering. For example,

$$d_1||||d_2d_3||d_4|||$$

corresponds to 9553. The only arrangement that yields an invalid solution is $|||||||||d_1d_2d_3d_4$, which corresponds to 0000. Thus, the number of valid combinations is yielded by stars and bars:

$$\binom{9+4-1}{4} - 1 = \frac{13\,(12)\,(11)\,(10)}{(4)\,(3)\,(2)\,(1)} - 1 = \boxed{714}$$

[Weakly Descending]

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Roundtable Handshakes

100 people sit around a roundtable. Each person raises either their left hand, right hand, or both hands, each with 1/3 probability. If two adjacent people raise their adjacent hands, they shake hands (e.g., a person who raises their right hand and a person to their right raises their left hand would shake hands). What is the expected number of handshakes?

SOLUTION:

Let H_i be the event that person i shakes hands with person $i+1$ (with person 100 shaking hands with person 1). Then the total number of handshakes is

$$H = \sum_{i=1}^{100} H_i$$

By linearity of expectation,

$$E(H) = \sum_{i=1}^{100} E(H_i)$$

Now we analyze H_i. The probability person i raises their right hand is $1/3 + 1/3 = 2/3$. The probability person $i+1$ raises their left hand is $1/3 + 1/3 = 2/3$. Thus, the probability that person i and person $i+1$ shake hands is $P(H_i) = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$. Therefore,

$$E(H) = \sum_{i=1}^{100} \frac{4}{9} = \boxed{\frac{400}{9}}$$

3.3 Coupon Collector Problem

3.4 Markov Chains

Rabbit Jump 'Til Termination

A rabbit starts at the origin on a number line. At each second, it jumps to the right with probability 1/3 and to the left with probability 2/3. The rabbit stops when it reaches -2, 2, or after 8 jumps, whichever comes first. What is the expected number of jumps?

SOLUTION 1:

Note that the jumps must terminate on an even second. With this in mind, we can calulate the probability it terminates at each even second. Denote e_n as the probability it termaintes on the n^{th} second.

e_2: LL or RR, so $e_2 = \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = \frac{5}{9}$
e_4: LR or RL, then probability it terminates in two seconds, or e_2. Thus $e_4 = 2\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)e_2 = \frac{20}{81}$.
e_6: We follow the same logic as above. $e_6 = 2\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)e_4 = \frac{80}{729}$
e_8: We must include ending at 0 in addition to -2 and 2.
$e_8 = 2\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)e_6 + \left(2\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)\right)^4 = \frac{64}{729}$

Thus,

$$E(X) = 2\left(\frac{5}{9}\right) + 4\left(\frac{20}{81}\right) + 6\left(\frac{80}{729}\right) + 8\left(\frac{64}{729}\right) = \boxed{\frac{2522}{729}}$$

SOLUTION 2:

We can use a "survival" probability to determine the expected value. Let e_n be the probability of a given rabbit not being terminated at time n. Again note that the rabbit can only terminate on an even second. Then,
$e_1 = 1$ $e_2 = 1$ $e_3 = \frac{4}{9}$ $e_4 = \frac{4}{9}$ $e_5 = \frac{16}{81}$ $e_6 = \frac{16}{81}$ $e_7 = \frac{64}{729}$ $e_8 = \frac{64}{729}$ Thus,

$$E(X) = \sum_{n=1}^{8} e_n = 1 + 1 + \frac{4}{9} + \frac{4}{9} + \frac{16}{81} + \frac{16}{81} + \frac{64}{729} + \frac{64}{729} = \boxed{\frac{2522}{729}}$$

3.5 Runs and Sequences

3.6 Random Walks

3.6.1 Gambler's Ruin

3.7 Advanced Probabalistic Thinking

Painting the Circumcircle

A circle is drawn. A first point is randomly placed on the circle, and then a second. Draw a green line clockwise from the first point to the second point. A third point is randomly placed on the circle, then a fourth. Draw a red line counterclockwise from the third point to the fourth point. What is the probability that the two line segments intersect?

SOLUTION:

Acute Inscribed Triangle

Three points are independently and randomly placed in a circle, and connected to form a triangle. What is the probability that the triangle is acute?

3.7.1 Reduced Spaces

2 Dice before 2 Dice

Roll a die until a 2 and a 3 are rolled at least once. What is the probability that exactly 2 6s appear?

SOLUTION:

3.8 Game Theory and Nim

CHAPTER 4

Advanced Counting Techniques

Appendices

APPENDIX A
Title

There once was a very smart but sadly blind duck. When it was still a small duckling it was renowned for its good vision. But sadly as the duck grew older it caught a sickness which caused its eyesight to worsen. It became so bad, that the duck couldn't read the notes it once took containing much of inline math just like its favoured equation: $d = u_c \cdot k$. Only displayed equations remained legible so it could still read

$$d = ra^k e.$$

That annoyed the smart duck, as it wasn't able to do its research any longer. It called for its underduckling and said: "Go, find me the best eye ducktor there is. He shall heal me from my disease!"